

Sooyong Shin

📍 Daejeon, South Korea ✉ exceldra5@gmail.com in exceldra5 🌐 exceldra5

Research Interests

I am interested in **Computer Vision and Robot Perception**, particularly in building **Spatial Intelligence** that enables robots to understand and interact with complex real-world environments. My research interests span **SLAM**, geometric and semantic scene understanding, and temporally persistent perception. I am particularly interested in integrating **Multimodal and Vision-Language Models** with spatial representations and memory to connect visual perception with context-aware reasoning and reliable robotic action.

Keywords: Robot Perception, Spatial Intelligence, SLAM, Vision-Language Models

Education

Korea Advanced Institute of Science and Technology	<i>Daejeon, South Korea</i>
<i>B.S. Candidate in Electrical Engineering and Computer Science (Double Major)</i>	<i>Feb 2018 – Feb 2027</i>
<i>English: TOEFL iBT 5.5/6.0 (MyBest Scores)</i>	

Publications

NavOCR: A Dataset Generator for Navigation-Relevant Text Detection in Mobile Robots
 Chaehyeuk Lee*, **Sooyong Shin***, Sehyeon Yoon*, Youngsun JAE*, Jinmyoung Lee, Seula Lee, Tu Vo, Sheir A. Zaheer, Chan Y. Park (*equal contribution)
NeurIPS 2026 Evaluations & Datasets Track (Poster) 🌐

Research Experience

KAIST NMAIL (Supervisor: Sungho Jo)	<i>Daejeon, South Korea</i>
<i>Research Intern</i>	<i>Mar 2026 – Jul 2026</i>
◦ GRF-based body weight estimation from walking time-series data	
KAIST URL (Supervisor: Hyun Myung)	<i>Daejeon, South Korea</i>
<i>Research Intern</i>	<i>Jun 2025 – Feb 2026</i>
◦ Conducted research on robot perception	
<i>Individual Study</i>	<i>Mar 2025 – Jun 2025</i>
◦ Conducted an individual study on ROS and visual SLAM	
KC-ML2	<i>Seoul, South Korea</i>
<i>Research Intern</i>	<i>Aug 2024 – Feb 2025</i>
◦ Conducted research on relocalization for robot mapping	
◦ Investigated text-based failure recovery in monocular visual SLAM 🌐	
◦ Skills: C++, Python, ORB-SLAM3, YOLO, PaddleOCR	
KAIST SIIT (Supervisor: Junmo Kim)	<i>Daejeon, South Korea</i>
<i>Individual Study</i>	<i>Jul 2024 – Aug 2024</i>
◦ Conducted an individual study on vision-language models	

Work Experiences

Bookend

Daejeon, South Korea

Software Engineer

May 2023 – Aug 2024

- Built the frontend and backend of **SENTENCIFY** [🔗](#)([Chrome Web Store](#) [🔗](#)), a multilingual AI writing and paraphrasing application integrating multiple large language model APIs
- Developed the service using **React**, **FastAPI**, and **MongoDB**

Software Engineering Intern (Frontend)

Dec 2022 – Feb 2023

- Developed **TYPHERE**, a React Native mobile platform connecting writers and publishers throughout the publishing workflow

Projects

CareON — Safe Medical AI for Post-Discharge AML Care

Jul 2026

Eccellence Award (2nd Place), Medical AI Competition 2026 [🔗](#) [🔗](#)

- Served as **AI & Systems**, designing a fail-closed medical AI system for post-discharge AML care.
- Integrated deterministic safety rules, grounded RAG, abstention, and audit logging to support reliable and traceable patient guidance.
- Built the **FastAPI** backend and **Flutter** client, and evaluated the system using synthetic clinical scenarios.

KAIST Swimming Team KAORI Web Platform [🔗](#) [www.kaist-kaori.org](#)

Since 2024

Full Stack Developer

- Developed and maintained a full-stack web platform for managing member profiles and swimming records
- Enabled individual and team performance analysis through structured historical record data
- Skills: React, FastAPI, MongoDB

Skills

Programming: Python, C++, JavaScript, TypeScript

AI & Robotics: PyTorch, ROS, ORB-SLAM3, YOLO, PaddleOCR

Development: React, React Native, FastAPI, MongoDB, Git

Leadership Experience

KAIST MR (Microbot Research) [🔗](#)

Daejeon, South Korea

Member

Since Mar 2025

AI Robotics KR [🔗](#) [🔗](#)

South Korea

Community Lead

Sep 2025 – Sep 2026

Community Manager

Oct 2024 – Present

Republic of Korea Army

Paju, South Korea

Sergeant; honorably discharged

Mar 2019 – Nov 2020

Korean Red Cross Lifeguard

Daejeon, South Korea

Since Aug 2018

KAIST Swimming Team KAORI [🔗](#)

Daejeon, South Korea

Senior Member

Since Mar 2018

President of KAORI

2022

Chair, KAIST Swimming Competition

2023